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Review

Application of support vector machine models for forecasting solar and wind energy resources: A review

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A B S T R A C T

Conventional fossil fuels are depleting daily due to the growing human population. Previous research has proved that renewable energy sources, especially solar and wind, can be suitable alternatives to the conventional energy sources that could satisfy global demand and protect the atmospheric environment. There are many factors that influence the performance of solar and wind energy predicting tools. The accurate forecasting of solar and wind energy resources is highly needed for the optimum utilization of these resources. Different methods have been applied to forecast solar and wind energy resources. Prediction performance of the support vector machine modeling approach found to be better than other modeling approaches. The support vector machine is fast, simple-to-use, reliable and provides accurate results. Findings based on critical analysis suggests that the hybrid support vector machine models can reach much higher accuracies than other models for both solar and wind energy predictions for most of the locations. This investigation highlighted main problems, opportunities and future work in this research area. Novel hybrid models are proposed for further investigation for more accurate predictions of solar and wind energy resources.

Contents

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1. Introduction

Energy plays a substantial role in modern society. Conventional fossil fuels are expected to be depleted due to growing demand and rapid industrialization. Renewable energy generation has drawn much attention from industries and researchers in recent decades mainly due to the abundance and sustainability of wind and solar energy.

In recent decades, many wind turbines and photovoltaic cells have been installed worldwide (Xu et al., 2018; Menezes et al., 2018). However, there are still issues, including large-range variable amount of power generation, due to variation in wind speed and direction. For solar energy, different parameters, such as solar elevation angle, haze effect and cloud cover, will cause fluctuations in output (Ren et al., 2015). The intermittent and variable output could lead to heavy negative impacts on grid, electricity transmission and distribution equipment, which prevent widespread use of green energy generation. Output forecasting has thus become important for the generation and implementation of wind and solar power systems.

Conventional and empirical models have been applied in a traditional way to forecast the resources of wind and solar energy, but they have demonstrated insufficient accuracy as well as other important limitations (Qazi et al., 2015; Lawan et al., 2017). Fortunately, artificial intelligence-based techniques have addressed these issues effectively. Artificial intelligence methodologies are relatively a new sort of modeling approach that have shown very promising results in modeling and analysis of systems in various branches of science. Such methodologies have the potential to deal with the uncertainties and other shortcomings of the traditional methods to real-world applications. Related applications will likely become more and more prevalent due to their robust, fast and accurate responses. There are many artificial intelligence forecasting models that have been developed for highly accurate estimations. Artificial neural network (ANN) is one of them that has been widely implemented to forecast system outputs in different investigations. However, many investigations have revealed disadvantages of ANNs. In other words, in some cases the prediction results of the ANNs are incorrect and/or consume too much time for a large neural network. Moreover, there is no proven method for selecting the numbers and sizes of hidden layers as well as activation functions to develop a high-precision model (Zendehboudi, 2016; Fayazi et al., 2014). Fortunately, another artificial intelligence forecasting model—support vector machine (SVM) approach has been confirmed to have better accuracy and speed in solving nonlinear problems.

This paper aims to provide a rigorous and critical analysis of the state-of-the-art review on the application of support vector machine models for solar and wind energy forecasting. No studies were found to provide a comprehensive analysis of this subject in the available literature. This paper is organized into five sections: after passing through introduction in Section 1, Section 2 presents the research methodology procedures and the steps to select the publications analyzed. Section 3 explains relevant descriptive statistics, while a brief introduction of SVMs is presented in Section 4. In Sections 5 and 6, developments of SVMs in the fields of solar and wind energy are presented by considering subgroups of application of SVMs. Next, the research gaps in these fields are identified and discussed, and finally Section 7 concludes this study. This work allows recognizing the main problems from the current available literature.

2. Research methodology

The SVM approach to predict wind speed was first introduced 14 years ago. More and more investigations have been conducted in this field and lots of advances have been achieved for the last few years. To assess the performance of SVM in the field of renewable energy, the authors have carried out a comprehensive review on the SVM models in wind and solar energy resources. By adopting a systematic literature review methodology, firstly, the publications in the desired areas are extracted and selected; and, secondly, the analysis of the publications was carried out by grouping them into different categories based on the SVM methodology application. Previous published papers on application of SVM in solar and wind energy were collected by searching through ScienceDirect, Engineering Village, ISI Web of Science, and Google Scholar databases. Further, at the end, a search in the databases of the major international publishers, such as Elsevier, IEEE Xplore, Springer, Taylor & Francis, ASME, Hindawi and Wiley, was conducted to guarantee the accuracy of gathered related papers. Keywords to select the studies were chosen taking into consideration the main words referring to this field and the words that widely adopted by scholars. In each research string, the keywords used for the selection of articles are: ‘solar’ or ‘wind’ or ‘estimation’ or ‘forecasting’ or ‘data-driven’ or ‘artificial intelligence techniques’ or ‘support vector machine’ or ‘SVM’ or ‘regression’ or ‘soft computing’ or ‘self-organizing maps’. The research was carried out by limiting the database and searching for keywords in “article title, abstract, keywords” and then adding constraints concerning “document type” (“article” “article in press” and “review”) and “subject area” focusing on research areas related to the solar and wind energy. Data were collected in December 2017. Different criteria were adopted and implemented to filter the articles. We focused on contributions published in English peer-reviewed academic journals without any restriction on the publication date. However, it should be highlighted that in the review process, conference papers, masters and doctoral dissertations, and the articles without clear information regarding the publisher and page number were excluded. Articles were selected for further analysis if the paper assessed, discussed or pointed out to forecasting solar and wind energy resources. After this preliminary step, the selected full papers were read independently to increase the effectiveness of the collection process. The reviewed literature is mostly published between 2009 and 2017 and mainly related to critical review of forecasting models. The forecasting statistical accuracies are compared systematically for both solar and wind resources for various modeling approaches. Different spatial correlation models, artificial intelligence methods and hybrid models are systematically compared and prediction performances are reported.

3. Descriptive statistics analysis

3.1. Distribution of publications across the period

In response to the rising interest in the fast and precise prediction of wind and solar energy resources, support vector machines have been frequently implemented and got wide acceptance since 2009 (Fig. 1). The selection process described in the previous section yielded a list of 75 publications. The interest in the use of support vector machine methodology is the significant recent advances in the development of the methodology itself. Although application of this kind of modeling is a relatively new area of research in estimation of solar and wind energy resources, we observed that the attention of scholars around the world regarding application of support vector machines is dramatically growing day by day in the fields of solar and wind energy. A large number of publications were found for the time period between 2015 and 2017 (47 articles). The increasing number of studies can be explained due to the fast and accurate outputs by such intelligent models as well as the new key objective in utilization of renewable energy resources.

3.2. Distribution of publications across the subject area

Fig. 2 presents the information about distribution of publications across the subject area for solar and wind energy. The 75 collected articles were divided according to the research directions in which they have been published. Based on the distribution of the articles, it is obvious that application of SVMs is discussed in different contexts and from many different perspectives in the solar and wind energy resources. Fig. 2 (a) indicates the results for solar energy that consists of 42 articles. The field with the most research is application of SVM for solar radiation (50%), followed by SVM for solar collector and photovoltaic systems (24%) and SVM for solar irradiation (21%). Fig. 2 (b) shows the results for wind energy that consists of 35 articles. From this figure, it was found that the majority of researchers implemented SVM approaches to forecast wind speed (69%), followed by wind power (29%). In contrast, application of SVMs in solar air heater systems (2%), solar insolation (2%) and wind direction (3%) received scant attention.

4. SVM modeling approach

An SVM is a machine learning algorithm based on statistical learning theory and the principle of structural risk minimization, which was presented firstly by Cortes and Vapnik in 1995 (Cortes and Vapnik, 1995). The network structure of an SVM can be seen in Fig. 3. SVMs have been successfully implemented for various purposes, such as images retrieval (Tao et al., 2006), fault diagnosis (Tian et al., 2015), text detection (Kim et al., 2001) and regression problems (Hemmati-Sarapardeh et al., 2014). The main idea of this approach is transforming the nonlinear input area to an area with high-dimensional properties to find a hyper-plane via nonlinear mapping. For classification, pattern recognition and analysis of regression, SVMs are mostly implemented and usually outperformed other methodologies such as traditional statistical models that have been developed earlier (Huang et al., 2002; Sung and Mukkamala, 2003). The SVR, support vector regression, is the SVM utilization for function approximation and regression. Different basic kernel functions are used in SVM models. The functions can be classified as polynomial (Poly), exponential radial basis function (ERBF), radial basis function (RBF), sigmoid and linear. A training dataset of input-output pairs is considered as $Z = \{\chi_i, \gamma_i | i = 1, 2, 3, \ldots, n\}$, where $\chi_i \, \varepsilon \, R^q$, q is the dimensional input vector, $\gamma_i \, \varepsilon \, R$ is the corresponding target value and n refers to the training data size. The regression model can be constructed, as shown in Equation (1):

$$\gamma = \omega^T \theta(\chi) + b \tag{1}$$

where ω is the weight vector, b is the bias term and $\theta(\chi)$ is representative of a nonlinear mapping function, which maps χ into higher dimensional feature space. To obtain ω, it is necessary to minimize the following regularized function, which can be formulated as in Equation (2), with the constraint of Equations (3-5):

$$min\left\{\frac{1}{2}\omega^2 + C\sum_{i=1}^{N}\left(\xi_i + \xi_i^{(*)}\right)\right\} \tag{2}$$

$$\gamma_i - \left\{\left(\omega^T\theta(\chi_i)\right) + b\right\} \leq \psi + \xi_i \quad i = 1, 2, \ldots, N \tag{3}$$

$$\xi_i, \xi_i^{(*)} \geq 0 \quad i = 1, 2, \ldots, N \tag{4}$$

where ψ is equivalent to the function approximation accuracy placed on the training data samples, $\xi_i^{(*)}$ and ξ_i represent the positive slack variables and C is the penalization parameter of the error that is applied to control the trade-off between the regularization term and empirical risk. Ultimately, the SVR is solved by introducing Lagrange multipliers, δ_i and δ_i^*, and exploiting the constraints, which has the following form:

$$f(\chi) = \sum_{i=1}^{N}\left(\delta_i - \delta_i^*\right)K(\chi, \chi_i) + b \tag{5}$$

Generally, the characteristics of the SVM method can be briefly stated as:

I. Considerably precise and robust,
II. Able to model complex nonlinear decision boundaries,
III. Less prone to over fitting in comparison with other models,
IV. Exhibit a compact description of the learned model,
V. Potential of implementation in pattern recognition, regression and classification.

In recent years, SVM modeling approaches have been

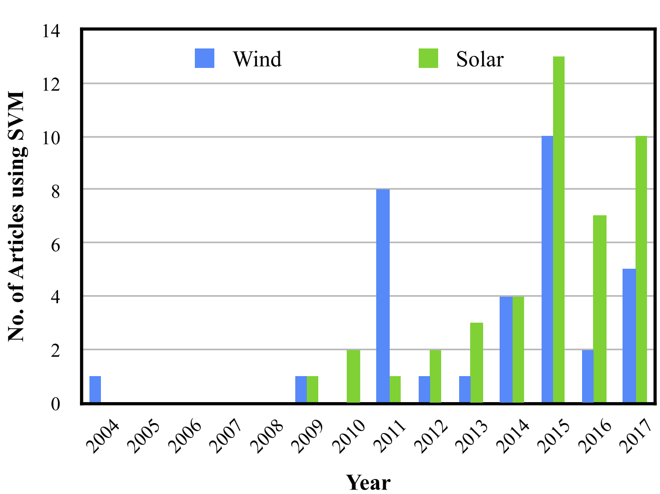

Fig. 1. Number of articles using SVMs as a function of published year.

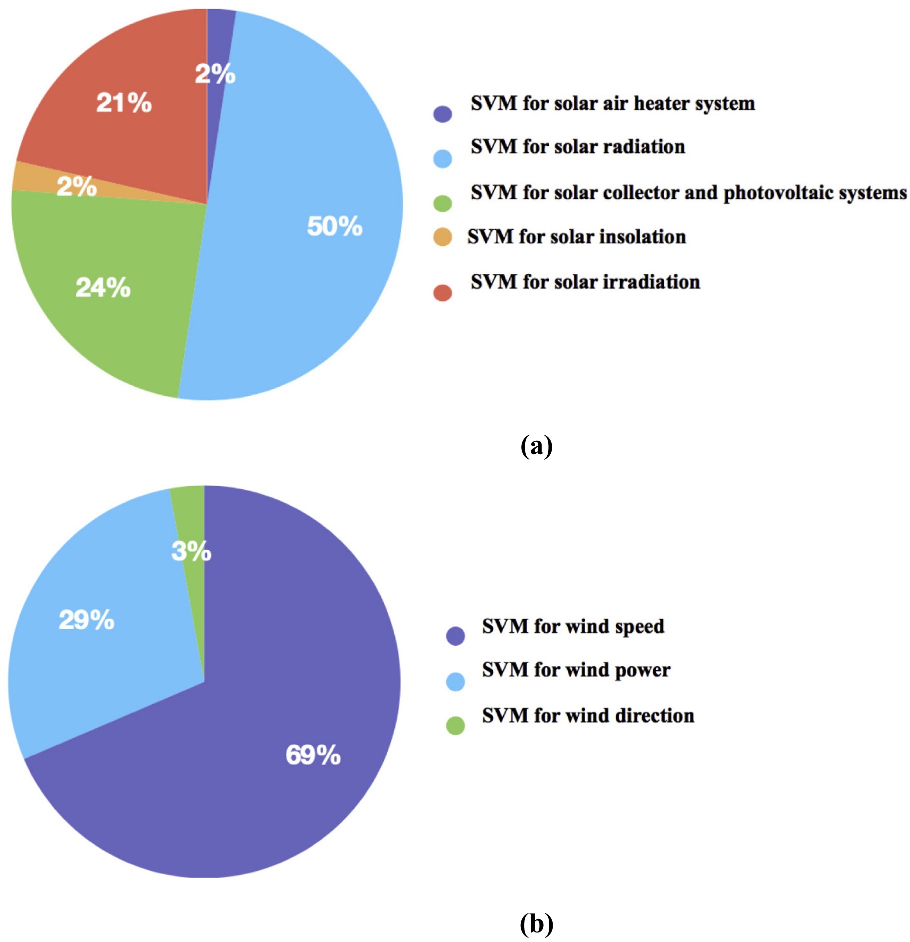

Fig. 2. Distribution of publications across the subject area for (a) solar energy; (b) wind energy.

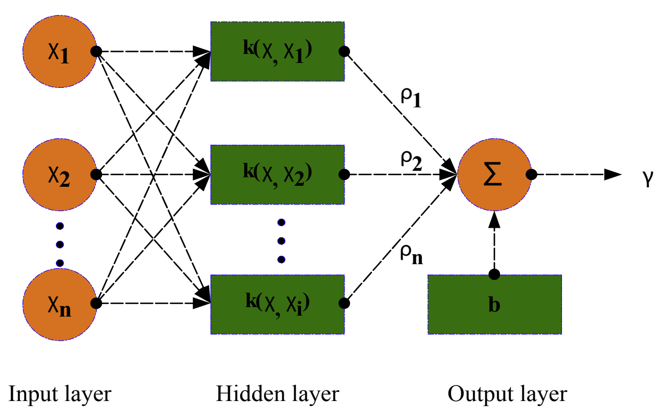

Fig. 3. Network structure of SVM.

implemented extensively in the fields of solar and wind energy and can range from single to hybrid and complex models. In general, they can be classified as shown in Table 1. The least square support vector machine (LSSVM) that was introduced by Suykens and Vandewalle (1999) is an extension of the standard SVM. A distinct feature of the LSSVM method that differs from the standard SVM is that a set of linear equation problems is solved instead of a convex quadratic programming, which makes the numerical implementation relatively easier (Suykens and Vandewalle, 1999). Indeed, the accuracy of a SVM is highly dependent on a precise selection of its parameters. In this regard, previous studies have sought to hybridize the SVM approaches with unique optimization techniques, such as grid search, firefly algorithm (FFA), genetic algorithm (GA), particle swarm optimization (PSO), cuckoo optimization algorithm (COA) and so on, because they have the capability to boost efficiency and accuracy as well as the speed of calculation in machine-learning technologies.

5. Development of SVM in solar energy

5.1. SVM for solar air heater system

As conventional methods could not be used easily to model a solar air heater system (SAHS) because of its multi-variable nature, Esen et al. (2009) used an LSSVM to evaluate the efficiency of a new SAHS. In the LSSVM model, seven variables (e.g., air temperature entering and leaving the collector unit, temperatures of the different points at the absorbing plate and solar radiation) were used as inputs. Collector efficiency was the only output variable. A comparison between the measurements at 0.03 kg/s and 0.05 kg/s air mass flow rates and estimations was carried out to determine the integrity of the model via root mean squared error (RMSE), coefficient of determination (R^2) and coefficient of variation (COV). It was found that the model had high capability in predicting SAHS, as presented in Table 2.

Table 1
SVM approaches in solar and wind energy.

Approaches	Area of research
Single Support Vector Machine, SVM	Wind & Solar
Least Square Support Vector Machines, LSSVM	Wind & Solar
SVM-Firefly Algorithm, FFA	Wind & Solar
SVM-Wavelet Transform, WT	Wind & Solar
SVR-Cuckoo Optimization Algorithm, COA	Wind & Solar
Support Vector Regression, SVR-Gradient Boosted Regression, GBR-Random Forest Regression, RFR	Solar
Seasonal Autoregressive Integrated Moving Average, SARIMA-SVM	Solar
SVM-Self-Organizing Maps, SOM-Particle Swarm Optimization, PSO	Solar
Evolutionary Seasonal Decomposition Least-Square, ESDLS-Support Vector Regression, SVR	Solar
K-means Clustering, KC-SVM- Genetic Algorithm, GA	Solar
Fuzzy Regression Function, FRF-SVM	Solar
Forward Regression, FR-Quadratic Kernel Support Vector Machine, QKSVM	Solar
WT-PSO-SVM	Solar
Kernel Support Vector Machine, KSVM-Group Regularized Estimation under Structural Hierarchy, GRESH	Solar
KSVM-Regularized Algorithm under Marginality Principle, RAMP	Solar
SVM-PSO	Wind
SARIMA-LSSVM	Wind
Fuzzy Group Support Vector Machine, GFSVM	Wind
Ensemble Empirical Mode Decomposition, EEMD-SVM	Wind
SVR-Unscented Kalman Filter, UKF	Wind
SVM-Grey Model, GM	Wind
ARIMA, Autoregressive Integrated Moving Average-ELM, Extreme Learning Machine-SVM-LSSVM	Wind
Reduced SVM, RSVM	Wind
C-LSSVM-PSO combined with Gravitational Search Algorithm, PSOGSA	Wind
Empirical Wavelet Transform, EWT-Coupled Simulated Annealing, CSA-LSSVM	Wind
Fast Ensemble Empirical Model Decomposition, FEEMD-Bat Algorithm, BA-LSSVM	Wind
Phase Space Reconstruction, PSR-SVR-GA	Wind
Piecewise Support Vector Machine, PSVM	Wind
SVR-Lavenberg–Marquardt Neural Network, LMNN-Broyden Fletcher Goldfar Shanno Neural Network, BFGSNN-Bayesian Regularization Neural Network, BRNN	Wind
LSSVM-GSA	Wind
WT-GA-SVM	Wind
SVM-Enhanced Markov Model, EMM	Wind
EEMD- Principal Component Analysis, PCA-LSSVM-Bat Algorithm, BA	Wind
Autoregressive Fractionally Integrated Moving Average, ARFIMA-LSSVM	Wind
Orthogonal Test, OT-SVM	Wind
Multi–Variable Regression, MVR-SVM	Wind

Table 2
Performance comparison of different kernel functions for 0.03 kg/s mass flow rate (Esen et al., 2009).

LSSVM models with optimum parameters	Statistical model validation parameters		
	RMSE	R^2	COV
LSSVM-RBF	0.0024	0.9997	2.1194
LSSVM-Poly	0.0044	0.9990	3.2285
LSSVM-linear	0.0061	0.9980	4.4861

5.2. SVM for solar radiation

Chen et al. (2011) used SVM models to forecast monthly mean daily solar radiation using air temperatures. A total of 28-year (1973–2000) monthly mean daily data samples were obtained from Chongqing meteorological station in China. In their research, seven combinations of air temperatures, namely; (1) T_{max}, (2) T_{min}, (3) T_{max} - T_{min}, (4) T_{max} and T_{min}, (5) T_{max} and T_{max} - T_{min}, (6) T_{min} and T_{max} - T_{min} and (7) T_{max}, T_{min} and T_{max} - T_{min}, were taken into account as input features for SVM models. Furthermore, three types of kernel functions, namely; linear, Poly and RBF, were studied. The developed SVM models were compared with several empirical temperature-based models. According to RMSE, relative RMSE (RRMSE), NSE and R^2, the suggested SVM model using T_{max} and T_{min} with Poly kernel function had better performance than other SVM and empirical methods.

Wu and Liu (2012) developed five SVM models with various input attributes to estimate monthly mean daily solar radiation by considering 13 years' meteorological data for twenty-four stations all over China. Furthermore, two empirical temperature-based methods were defined to be compared with the developed SVMs. According to RMSE, Nash-Sutcliffe (NSE) and coefficient of residual mass (CRM), SVMs outperformed the empirical models. The authors concluded that the SVM methodology may be a promising alternative to the traditional approach for forecasting solar radiation at any location for which air temperatures are available.

SVM has been successfully used to address many forecasting problems. Chen et al. (2013) used an SVM to estimate daily solar radiation based on the duration of exposure to sunshine in Liaoning province in China. In this paper, application of SVM models using different input attributes versus empirical sunshine-based models was presented. Numerous statistical criteria were used to examine reliability of the method. Generally, the SVM models presented good performances and significantly outperformed the empirical models. However, the developed SVM model using sunshine ratio as input attribute was preferred due to its greater accuracy and simplicity.

Ramedani et al. (2014) used a fuzzy linear regression (FLR) and an SVM to forecast global solar radiation (GSR) in Tehran province in Iran. Various meteorological data from the studied region were selected as the model inputs while GSR was selected as the model output. Performance of the proposed models was evaluated based on RMSE and correlation coefficient (R). The results showed that SVM with RBF kernel function yielded promising results for the prediction of GSR in comparison with FLR model.

Chen and Li (2014) developed twenty SVM models to predict solar radiation via measured meteorological variables. In this investigation, different parameters, including different combinations of sunshine ratio, maximum and minimum air temperature, relative humidity and atmospheric water vapor pressure, were used as inputs that were collected from 15 stations in China. An analysis that compared SVM models with empirical models showed that SVM models achieved 14% greater accuracy. The authors recommended that the SVM modeling approach could be a suitable alternative for the traditional ways to forecast solar radiation.

Olatomiwa et al. (2015a) presented a hybrid SVM-FFA approach to forecast monthly mean horizontal GSR based on three meteorological variables for three different places in Nigeria. The authors utilized 252 data samples from 1987 to 2007 to develop the proposed model. Input parameters in this study were considered as sunshine duration, maximum temperature and minimum temperature. In this analysis, the hybrid model performance was compared against ANN and genetic programming (GP) models. To examine the ability and robustness of the suggested method, various statistical error tests, namely; RMSE, R^2, R and mean average percentage error (MAPE), were utilized. The comparison results indicated that the proposed model was more accurate for estimation of horizontal GSR, as shown in Table 3.

Chen et al. (2015) evaluated the transferability of SVM for the estimation of solar radiation in a subtropical zone in China. In this

Table 3
Summary of performance statistics of the SVM–FFA model against other methodologies for Nigeria (Olatomiwa et al., 2015a).

		RMSE	R^2	R	MAPE
SVM–FFA	Training	0.6988	0.8024	0.8956	6.1768
	Testing	1.8661	0.5300	0.7280	11.5192
ANN	Training	0.7673	0.7596	0.8730	6.7813
	Testing	2.0458	0.4659	0.6496	13.4305
GP	Training	0.7507	0.7619	0.8708	6.9594
	Testing	1.9532	0.5181	0.6899	13.2089

Table 4
Performance statistics of SVR model compared to other methodologies (Olatomiwa et al., 2015b).

	RMSE	R^2
SVR-Poly		
Training	1.363988	0.7703
Testing	1.510218	0.7395
SVR-RBF		
Training	1.141814	0.8425
Testing	1.905994	0.5877
ANFIS-ACO		
Training	1.3505	0.7748
Testing	1.5485	0.7313
ANFIS-DE		
Training	1.3503	0.7748
Testing	1.601	0.6926
ANFIS-GA		
Training	1.2769	0.7987
Testing	1.7696	0.6354
ANFIS-PSO		
Training	1.1157	0.8463
Testing	1.8015	0.5666
ANFIS		
Training	1.0699	0.8586
Testing	2.0628	0.5468

survey, fifteen years' monthly data from 1994 to 2008, including air temperature and long-term radiation, were collected from 32 meteorological stations in order to develop and depict integrity of the model. The performance evaluation of the model via R^2, RMSE and MAPE as well as *t*-test were carried out, which demonstrated the capability and reliability of the aforementioned model.

Mohammadi et al. (2015a) used two SVRs of RBF and Poly to forecast GSR on a horizontal surface, considering Isfahan province in Iran as the case study. The input parameters in this paper were sunshine hours and maximum possible sunshine hours. The models were developed via historical data that were collected from Iranian Meteorological Organization in the period of 1985–1991 and 1998–2003. Various statistical indicators, such as relative percentage error (RPE), MAPE, mean absolute bias error (MABE), RMSE, RRMSE and R^2, were employed to assess model performance and reliability. A comparison that was conducted between SVM models and empirical sunshine duration-based models yielded a satisfactory result with utilization of SVM-RBF for both daily and monthly mean scales.

Olatomiwa et al. (2015b) explored the potential of SVM approach for solar radiation prediction in Nigeria. In this paper, the authors decided to consider inputs as monthly mean maximum temperature, monthly mean minimum temperature and monthly mean sunshine duration. Poly and RBF were used as the SVR kernel functions to estimate solar radiation. The dataset for the needs of this investigation was accumulated from a meteorological station located at Iseyin, southwest Nigeria. The result of forecasting between SVRs and the other models was compared via RMSE and R^2, which depicted that SVR with Poly was more suitable, as shown in Table 4.

Mohammadi et al. (2015b) predicted horizontal GSR using a hybrid SVM-wavelet transform (SVM-WT) approach based on both daily and monthly mean scales for Bandar Abbas city in Iran. The input parameters were various combinations of the following factors: relative sunshine duration, difference between maximum and minimum ambient temperatures, relative humidity, water vapor pressure, average ambient temperature and extraterrestrial GSR on a horizontal surface. A dataset during the period 1992–2001 was considered for training of the model and the model was validated using the dataset from 2002 to 2005. After utilizing several statistical indicators and comparisons with other models, it was concluded that SVM-WT had favorable potential for forecasting global solar radiation (daily and monthly mean scales).

The application of SVM and ANN models for forecasting solar radiation on a tilted surface for Jeddah and Qassim locations in Saudi Arabia was investigated by Ramli et al. (2015). The collected data samples from the year 2002 from National Renewable Energy Laboratory (NREL) website were utilized to exhibit ability and integrity of the models. According to RMSE, the coefficient of correlation (CC) and the magnitude of relative error as well as the speed of computation for the algorithms, the SVM approach was considerably more accurate during computation and rapid in forecasting radiation on the tilted surfaces.

Piri et al. (2015) proposed an SVM using both RBF and Poly as kernel functions of the considered model in order to forecast solar radiation on the Earth based on data from Zahedan and Bojnoord cities in Iran. Various parameters, such as daily mean maximum temperature, daily mean minimum temperature, daily mean relative humidity and daily mean sunshine duration hours, were considered as inputs and the models were evaluated by RMSE and R^2. Generally, it was indicated that SVR models had higher capability than empirical models for prediction of solar radiation. The SVR with Poly approach was preferred for Bojnoord station, while the SVR technique with RBF was recommended to estimate solar radiation for Zahedan station.

Two hybrid modeling approaches, SVM-FFA and SVM-WT using RBF kernel function, were implemented by Mohammadi et al. (2015c) to predict monthly mean daily horizontal GSR. Two sets of meteorological parameters were used as inputs: (1) relative sunshine duration and (2) relative sunshine duration, air temperature difference, average air temperature and relative humidity. In this study, Shiraz city in Iran was considered as the case study. Data were collected for the period of 10 years for this city at geographical location of 27°02′N and 31°42′N as well as 50°42′E and 55°38′E. The accuracy of the models was assessed by considering various statistical errors, such as MAPE, MABE, RMSE, RRMSE and R^2. The results demonstrated how SVM-WT could be more effective and robust for predicting monthly mean daily horizontal GSR than SVM-FFA.

Deo et al. (2016) developed an SVM-WT model to predict incident solar radiation based on sixteen months of data (01-March-2014 to 30-June-2015). The data samples were collected from Brisbane, Cairns Aero and Townsville Aero stations. The input vectors for development of this model were sunshine hours, minimum temperature, maximum temperature, wind speed, evaporation and precipitation. A statistical comparison between the hybrid model and a single SVM indicated that the SVM-WT model performed better than the classical model in the short- and long-term evaluations.

Kim et al. (2016) debated about application of SVM-WT in prediction of daily solar radiation in two stations of Champaign (latitude, 40.0840° N; longitude, 88.2404° W; altitude, 219 m) and Springfield (latitude, 39.7273° N; longitude, 89.6106°W; altitude, 177 m) in Illinois. The data samples between 2007 and 2010 were considered for training and the model performance was tested using collected data in 2011. By considering different criteria, it was found that the application of SVM with wavelet decomposition is more robust to present accurate results.

The SVM-WT algorithm was applied by Shamshirband et al. (2016) to forecast daily horizontal diffuse solar radiation by taking clearness index as the only input parameter. Kerman city in Iran with the geographical location of 30°29′ N and 57°06′ E and the elevation of 1756 m above the sea level was considered for evaluation. The data samples between 2006 and 2012 were used to develop the model. A comparison between the proposed model in this paper with SVM-RBF, ANN and an empirical model via MABE, RMSE and R indicated that SVM-WT had better ability to forecast diffuse solar radiation, as shown in Table 5.

A hybrid machine learning method, including SVR, gradient boosted regression and random forest regression, was developed by Gala et al. (2016) in order to forecast solar radiation near the Spanish cities of A Coruña, Alicante, Almeria, Oviedo-Asturias, Barcelona, Ciudad Real and Salamanca. As a result, it was found that the hybrid model had superior ability for solar radiation prediction in these areas.

Liu et al. (2017a) proposed a two-step radiation zoning methodology using k-means cluster and SVM-GA according to: (a) global solar radiation, sunshine duration, temperature and relative humidity from 98 solar radiation observation stations and (b) sunshine duration, temperature and relative humidity from 562 stations without radiation. The suggested method was capable to combine the solar radiation observation stations and the stations without radiation in the process of classification.

Baser and Demirhan (2017) focused on the prediction of yearly mean daily horizontal global solar radiation using a hybrid approach that combined fuzzy regression functions with support vector machine (FRF-SVM). The authors compared the prediction robustness of the suggested FRF-SVM approach against an adaptive neuro-fuzzy system and a coplot supported-genetic programming model. The result of this study showed that the FRF-SVM method with a Gaussian kernel function outperformed the other models in estimating the horizontal global solar radiation over a dataset that was collected in Turkey.

Jiang and Dong (2017a) introduced forward regression on the quadratic kernel support vector machine (QKSVM-FR) for building a quadratic regression method using forward regression to choose the most important input variables for estimating the global horizontal radiation in the Tibet Autonomous Region. It was indicated that the method was superior forecasting in the modeling against the other benchmark models.

Quej et al. (2017) evaluated performance of three different artificial intelligence techniques, such as adaptive neuro fuzzy inference system (ANFIS), ANN and SVM, for forecasting daily horizontal global solar radiation in Yucatán Peninsula, México. Model capability was assessed with statistical error tests, such as RMSE, MAE and R^2. The performance evaluation highlighted that SVM approach with requirements of daily maximum and minimum air temperature, extraterrestrial solar radiation and rainfall had better performance than the other models and might be an interesting alternative to the usual methodologies for estimating solar radiation.

Table 5
The attained MABE, RMSE and R for all models considering testing dataset (Shamshirband et al., 2016).

Model	MABE	RMSE	R
SVM−WT	0.5757	0.6940	0.9631
SVM−RBF	1.0877	1.2583	0.8599
ANN	1.1267	1.3184	0.8392
Empirical model	1.2171	1.4548	0.8156

5.3. SVM for solar collector and photovoltaic systems

Varol et al. (2010) developed three models, namely; ANN, ANFIS and SVM, to forecast performance of a solar collector system using sodium carbonate decahydrate ($Na_2CO_3.10H_20$) as phase change material (PCM). The dataset was earned by doing experiment in Turkey in March 2003 in order to figure out integrity of the proposed models. The comparison between performance of SVM, ANN and ANFIS models via R^2 proved SVM as a reliable and novel modeling approach for the application of interest, as illustrated in Table 6.

The estimation of parabolic trough solar collector system using an LSSVM with RBF kernel function has been carried out by Liu et al. (2012). The required data for training of the model were obtained from both simulation and experimental tests. The solar flux, inlet temperature and flow rate of heat transfer fluid (HTF) as well as outlet temperature of the HTF were considered as inputs variables. On the other hand, the solar collector efficiency was considered as the output parameter. The model was found able to propose superior forecasting in modeling of solar collector systems.

Fonseca et al. (2012) developed an SVM to estimate power output of a 1-MW photovoltaic power plant for one year in Kitakyushu, Japan. Meanwhile, numerically predicted cloudiness was implemented in order to evaluate its influence on the forecasting. To estimate the power production of the PV power plant, six variables, including normalized temperature, relative humidity, low level cloudiness, mid-level cloudiness, upper level cloudiness and extraterrestrial insolation, were applied as inputs vectors in this method. Predicted values using SVM and those earned by a persistence method were compared via RMSE, mean absolute error (MAE) and MAPE indicators. The results showed that the use of SVM methodology yielded predictions of power production with a good level of accuracy.

The estimation of short-term solar power was carried out by Zeng and Qiao (2013) using LSSVM. The input parameters used in this model were historical data of atmospheric transmissivity in a novel two-dimensional form as well as sky cover, relative humidity and wind speed, which were recorded between 1991 and 2005 at 1454 locations in the United States. The results indicated that proposed approach was more robust for estimation of short-term solar power than a radial basis function neural network (RBFNN) and reference autoregressive (AR) models, as illustrated in Fig. 4.

A hybrid model by combining SVM and seasonal autoregressive integrated moving average (SARIMA), namely; SARIMA-SVM, was developed by Bouzerdoum et al. (2013) for short-term power forecasting of a small-scale grid-connected photovoltaic (GCPV) plant. Experimental data samples of power produced by a small-scale 20 kW_p GCPV plant installed in Trieste in Italy were utilized to evaluate the performance of developed approach against those of single SVM and SARIMA approaches. The result was assessed via

Table 6
R^2 values using different test parameters for SVM, ANFIS and ANN (Varol et al., 2010).

Date of experiments	Test parameter	R^2 SVM ANFIS ANN		
		SVM	ANFIS	ANN
03.27.03 (PCM)	Q_u	0.992	0.910	0.876
	η	0.987	0.905	0.872
03.28.03 (PCM)	Q_u	0.995	0.942	0.899
	η	0.992	0.956	0.894
03.27.03	Q_u	0.982	0.934	0.832
	η	0.974	0.879	0.844

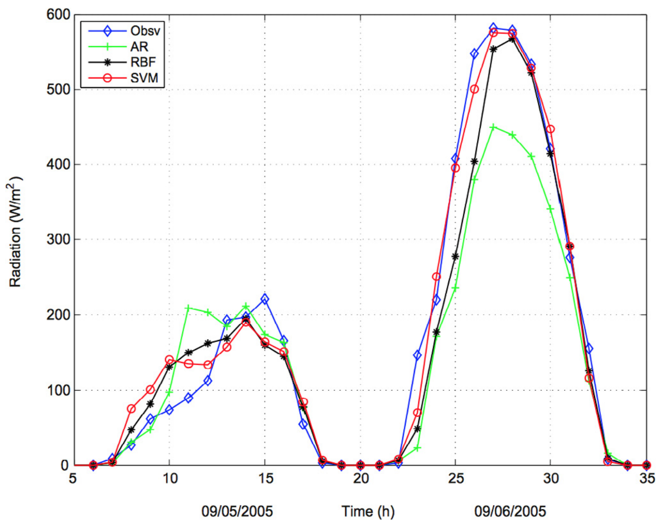

Fig. 4. Comparison of the 1-h-ahead predicted values of the three models with the observations in two consecutive days in Seattle (Zeng and Qiao, 2013).

R, normalized root mean square error (NRMSE), normalized mean bias error (NMBE) and mean percentage error (MPE), which showed the proposed model had better performance than the other benchmark methods.

Lin and Pai (2016) developed a hybrid model for forecasting monthly solar power output based on a dataset from Taiwan Power Company. The authors used LSSVM and seasonal decomposition to defined ESD-LSSVR model and used GA to optimize the parameters of LSSVM. The comparison between the aforementioned model performance and autoregressive integrated moving average (ARIMA), SARIMA, generalized regression neural network and LSSVR models via RMSE and MAPE illustrated the ability and superiority of the proposed hybrid model.

Three SVM models, namely; SVM-WT, SVM-FFA and SVM-RBF, were developed by Mojumder et al. (2016) to predict electrical and thermal performance in PV/T system. The comparative investigations between these three models were carried out by utilizing a dataset that was earned from experiments in Malaysia. The result of estimation was evaluated via RMSE, R^2 and R. The comparison depicted integrity and robustness of applying SVM-WT approach as an accurate model for electrical and thermal performance in PV/T systems versus other artificial intelligence models.

Assouline et al. (2017) explored the potential of SVMs and geographic information systems to predict the rooftop solar PV potential for the urban areas at the commune level (the smallest administrative division) in Switzerland. Following a 6-fold cross validation, the RMSE was utilized to evaluate the accuracy of different SVM methods. The results indicated that, on average, 81% of the total ground floor area of each building corresponds to the available roof area for the PV installation.

Ferlito et al. (2017) compared effectiveness of multiple linear regression, classification and regression tree, model tree M5, extreme learning machines, weighted k-nearest neighbours, multivariate adaptive regression spline, support vector machines, bayesian regularized neural networks, random forests, cubist and extreme gradient boosting in obtaining 12 h ahead prediction of PV power. A dataset of about seven years of 1 kW_p GCPV plant was considered. The considered models were compared applying two different training methodologies (online and offline) to identify the most performing training mode. The SVM was among the models, which was able to assure minimum prediction errors.

Eseye et al. (2017) developed a hybrid forecasting methodology that combines wavelet transform, particle swarm optimization and support vector machine (WT-PSO-SVM) for estimating short-term (one-day-ahead) generation power of a real microgrid PV system that was installed in Beijing in China. In this method, the wavelet was used to have a considerable impact on ill-behaved meteorological and supervisory control and data acquisition data. The daily MAPE and NMAE had 4.22% and 0.4% average values, respectively, outperforming seven other predictive models while the mean computational time was smaller than 15 s.

5.4. SVM for solar insolation

An investigation based on LSSVM was conducted by Ekici (2014) to predict the next day solar insolation. In order to train the proposed model, 1096 data samples measured by Turkish State Meteorological Service from 2000 to 2002 were considered and 365 data of next year, i.e. 2003, were used to test the proposed model. Numbers of the days from January 1, daily mean temperature, daily maximum temperature, sunshine duration and the solar insolation of the day before parameters was considered as input vectors while the daily solar insolation was the only output vector. Performance of the model was evaluated through several statistical indicators, such as RMSE, mean relative error (MRE), mean error function (MEF), R^2 and the coefficient of variance based on root mean square error (CVRMSE). According to the results, it was found that the model had favorable potential for forecasting the desired output.

5.5. SVM for solar irradiation

The SVM was applied by Cheng et al. (2014) to forecast short-term solar irradiance. Data were collected for one month at a costal site in Taiwan in order to develop a model and figure out its ability. The result of prediction was evaluated via root mean square percentage error (RMSPE) and MAPE, which indicated the model was highly capable in forecasting of such problems.

In 2015, Antonanzas-Torres et al. (2015) developed an SVM approach using RBF kernel function to forecast solar irradiation based on a case study in Spain. The daily basis data were collected from 14 meteorological stations by Spanish Advisory Service for Farmers to develop and subsequently evaluate integrity of aforementioned model. After forecasting the model produced reduction in MAE 41.4% and 19.9% as compared to classic parametric models.

Antonanzas et al. (2015) considered Spain as the case study and used a combination of geostatistical interpolation techniques with SVMs to map daily global irradiation based on temperatures, rainfall, humidity and wind speed. The authors used a dataset from 2000 to 2012, including 13388 data samples, to train the model and 1177 data samples of 2013 to test the model. A comparison between the literature using different soft computing techniques, such as ANNs, regression trees M5P, classic parametric models and extreme learning machines (ELM), proved the capability, robustness and reliability of SVM for prediction of solar irradiation.

Dong et al. (2015) used a hybrid model based on self-organizing maps, SVR and PSO to estimate hourly solar irradiance time series in both Colorado in USA and Singapore during day time, from 04:00 to 20:00 in Colorado and 07:00 to 19:00 in Singapore. The data samples had been collected during the period 1997 to 2013 for Colorado and during the period 2010 to 2013 for Singapore. NRMSE and NMBE were employed to offer a thorough comparative analysis for identifying the merits of the used model. Comparison between the hybrid model and the other popular statistical time serie models like ARIMA, linear exponential smoothing (LES), simple exponential smoothing (SES) and random walk (RW) was conducted, which indicated superior forecasting accuracy of the proposed model.

Urraca et al. (2016), in 2016, used two machine learning techniques, random forests and SVR, and the classical linear regression to forecast solar irradiation for horizons of 1 h in a site of Southeast Spain with geographical characteristics of 39°11′38″N and 0°26′13″W. The study involved the use of two approaches: fixed and moving models. Comparison showed high precision and effectiveness of applying SVM approach as a sufficient model for predicting solar irradiation versus other models.

The Hidden Markov model and SVM regression using RBF kernel function were applied by Li et al. (2016) for forecasting short-term solar irradiance of a photovoltaic system under various weather conditions. Comparison of experimental samples that were accumulated from Australian Bureau of Meteorology and the those of estimated by the offered models proved accuracy of SVM over 90%. In conducted analysis, it was indicated the models are able to predict future 5–30 min solar irradiance under different weather conditions.

Jiang and Dong (2017b) studied structured variable selection in kernel SVM-based approaches to estimate global horizontal irradiance. The kernel support vector machine-group regularized estimation under structural hierarchy (KSVM-GRESH) and kernel support vector machine-regularized algorithm under marginality principle (KSVM-RAMP) approaches were established. Selective cross validation-risk inflation criterion was applied to determine the regularization parameters in KSVM-GRESH and extended bayesian information criterion was implemented to select the parameters in KSVM-RAMP. The estimation results have shown the superior forecasting performances of KSVM-GRESH and KSVM-RAMP models based on the four real-world case studies in China.

Silva et al. (2017) reported the potential of Angstrom-Prescott model, SVM and ANN for predicting the daily global solar irradiation. The authors collected the experimental data points from 1996 to 2011 in Botucatu in Brazil. Different combinations of input variables were selected and different models were developed. Review of this article indicated an excellent agreement between the developed optimal SVM model results and the experimental data samples. It was found that the SVM method had better performance in modeling daily global solar irradiation in Botucatu in Brazil.

Bae et al. (2017) proposed a predictive scheme for hourly modeling of solar irradiance based on the weather classification and the SVM regression. A k-means clustering algorithm was applied to classifying the collected meteorological data, including the cloud cover. The researchers compared the results with those of estimated by ANN and Nar models using different statistical criteria. The results illustrated that the suggested SVM regression scheme significantly enhanced the estimation accuracy and reduced the energy storage system installation capacity.

6. Development of SVM in wind energy

6.1. SVM for wind speed

Mohandes et al. (2004) developed an SVM to predict wind speed. A dataset of 12 years between 1970 and 1982 for Madina city in Saudi Arabia was collected to figure out how the aforementioned model can be reliable and have satisfactory precision. A comparison between the results of SVM and multilayer perceptron (MLP) neural network models was carried out and it was indicated that mean square error (MSE) value of SVM model was lower than multilayer perceptron, as shown in Fig. 5.

The SVM modeling approach to forecast short-term wind speed was developed by Sreelakshmi and Ramakanthkumar (2008). To develop this model, a dataset of 10 years was utilized, which were collected at a weather station that had measured every 10 min. Review of this article indicated that the errors in predictions vary with the type of kernel, Epsilon and number of inputs. Furthermore, it was concluded that performance of the developed SVM highly outperformed ANN approach in terms of computational times.

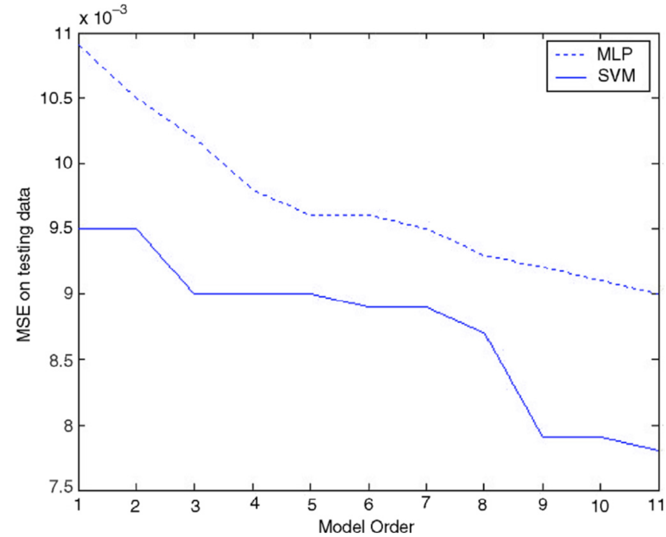

Fig. 5. Comparison between the MSE of SVM and MLP on testing data (Mohandes et al., 2004).

Zhou et al. (2011) estimated short-term wind speed using LSSVM in North Dakota based on the hourly wind speed data of year 2002. In their analysis, the dataset was divided into four seasonal datasets and huge attempts were made to present fine tuning of LSSVM model parameters by taking three types of linear, RBF and Poly kernel functions. The effectiveness of the models was evaluated via RMSE and showed robustness of LSSVMs in comparison with persistence model.

Ortiz-García et al. (2011) discussed about several new training structures based on SVR for a problem in wind speed prediction using dataset of five wind turbines in southeast Spain. The inputs were defined as the wind speed series at two grid points surrounding a park, wind direction and temperature at one of these points. The comparison results between the proposed models and models based on multi-layer perceptron showed robustness of SVRs.

Zhang et al. (2011) developed an SVM for forecasting wind speed. Performance of three different kernel functions was evaluated and PSO was selected as optimizer to determine the most effective solutions and tuning parameters. After comparing the results, it was concluded that utilizing RBF kernel function and PSO was more desirable. The authors illustrated that accuracy rate for forecasting wind speed could reach 98.667% using the developed model.

Salcedo-Sanz et al. (2011) hybridized SVM by evolutionary programming (EP) and PSO algorithms to evaluate performance of two evolutionary approaches in hyper parameters calculation. Experimental data samples from January to June 2006 of a wind farm in Spain were applied to estimate wind speed and figure out the robustness of the models. RBF was selected as kernel function of SVM and 80% of the data were used for training and the remaining 20% put aside to test the aforementioned models. MAE statistical quality measure parameter was used to evaluate performance of the models, and it was found that PSO and EP had favorable capability for the case of study.

Huang et al. (2011) studied applicability of LSSVM with GA to forecast short-term wind speed using a dataset that was measured from a wind farm located in Penghu in Taiwan. The input parameters were considered as the time, temperature, humidity and average regional wind speed. RBF was selected as the kernel function of the model. They found that the short-term wind speed considering GA-LSSVM could be predicted with the average error around 2.27%. At the end, the authors mentioned that the accuracy could be increased by considering more factors that affect the wind speed.

Short-term wind speed was predicted based on best wavelet packet decomposition method and SVM by Zeng et al. (2011). The data used were 1000 points that were collected in the duration of one month. 900 data samples were used for training and the remaining data were used to test the accuracy of the proposed model. The values predicted by this model demonstrated a good agreement with corresponding experimental data. The average absolute error (AAE) and ARE were found to be 0.3235 m/s and 5.49% of real data, respectively, as shown in Fig. 6.

Guo et al. (2011) developed a novel hybrid model for mean monthly wind speed prediction in Hexi Corridor in China. The model was composed of SARIMA and LSSVM, called SARIMA-LSSVM. The simulation results based on a dataset from January 2001 to December 2006 proved that the aforementioned model had higher efficiency and precision as compared against single ARIMA, SARIMA, LSSVM and ARIMA-SVM models.

Zhang et al. (2012) have integrated groups of models based on LSSVM into an aggregated model by using fuzzy theory. Three groups of LSSVM estimating models, such as univariate LSSVM, ARIMA-LSSVM and multivariate LSSVM, were constructed using a dataset from 1 January 2011 to 31 December 2011, which were collected from the Changshun wind park in Huade County, Inner Mongolia Autonomous Region in China. The prediction results of this group-forecasting model indicated robustness and reliability of the model for wind power prediction in the studied region.

In the following year, Hu et al. (2013) developed a hybrid modeling approach based on SVM, namely; EEMD-SVM, in order to boost the quality of wind speed forecasting model. The model predictability was evaluated by forecasting the mean monthly wind speed of three wind farms located in the northwest China. MAE and MAPE indices were employed to assess the model estimations in comparison with various modeling approach, such as ARIMA, SARIMA, SVM and EMD-SVM. Results showed that the suggested model performed well in terms of accuracy and consistency.

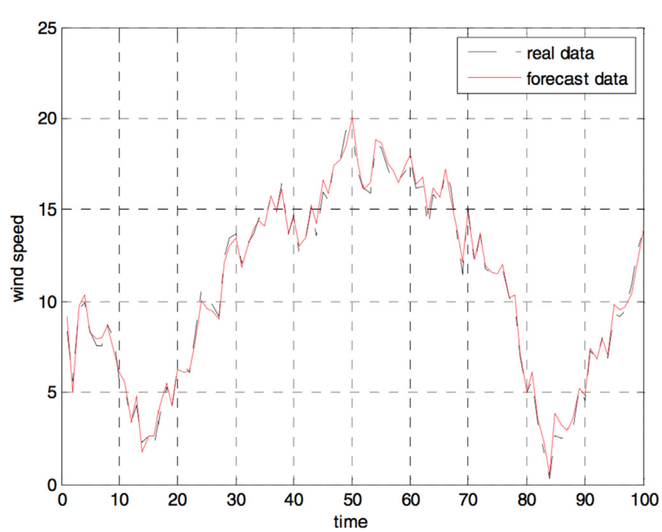

Fig. 6. Result of the short-term wind speed prediction (Zeng et al., 2011).

The SVR-UKF approach was proposed by Chen and Yu (2014) to forecast short-term wind speed. RBF was selected as kernel function of SVR model in this study. The SVR-UKF model was developed and validated using 700 wind speed data samples from three different sites within Massachusetts: Blandford, Chester and Falmouth. The data samples were measured every 10 min at the same position in each site and 25 m above the ground level, which were collected from Energy Efficiency and Renewable Energy at University of Massachusetts. Utilizing RMSE, MAPE and R^2 indices, the tested SVR-UKF indicated better accuracy in against ANNs, single SVR as well as autoregressive and autoregressive integrated with Kalman filter approaches. Therefore, SVR-UKF method was introduced for estimating both one-step-ahead and multi-step-ahead wind speed in these regions.

Liu et al. (2014) developed an SVM-WT approach integrated with GA to forecast short-term wind speed based on a case study of a wind farm in north of China. 1440 data samples were measured every 30 min in September 2012. Two other models, i.e. persistent method and SVM-GA, were also developed in order to show the reliability of the proposed model. According to MAE, MAPE and RMSE, the results showed that SVM-WT-GA method was an efficient method for forecasting short-term wind speed.

A hybrid model for wind speed prediction based on Grey theory and SVM was proposed by Xiao et al. (2014). MAPE and MSE were utilized to assess the model using a dataset of a wind turbine in Harlan in USA. Based on the results drawn from statistical analysis, the implementation of hybrid model could lead to better predictions for the studied station.

A hybrid model based on combination of the ARIMA, ELM, SVM and LSSVM using a Gaussian process regression model was developed by Wang and Hu (2015) to predict short-term wind speed based on a dataset from two wind farms in China. The data to train and test the aforementioned model include 576 mean 15-min wind speed observations from site 1 and 576 mean 30-min wind speed observations from site 2. The results based on MAE, RMSE and MAPE statistical criteria indicated that the developed method would be a promising alternative over the traditional approaches.

Kong et al. (2015) developed reduced SVM (RSVM) based on the concept of SVM, which was optimized by PSO algorithm to forecast wind speed in Neimenggu in China. The principal component analysis was applied and four models with different input combinations were defined. Comparing these models via RMSE, MAE and MAPE statistical criteria indicated that the 3-RSVM model, taking inputs as the wind speed, temperature and air pressure outperformed the other models, as shown in Table 7. Also, the comparison between aforementioned model and traditional SVM indicated the advantage in terms of computational time.

Wang, Y. et al. (2015) applied a hybrid model, C-LSSVM-particle swarm optimization with gravitational search algorithm (PSOGSA), based on LSSVM and Markov model to precisely estimate daily mean wind speed and 10 min average wind speed. LSSVM model optimized via PSO combined with GSA and four datasets of wind farms in northwest China, including 1096 samples, were used to develop the model. Performance of the model was assessed using various statistical error tests, such as RMSE, MAE and MAPE, and the results were compared against the other models. It was

Table 7
Wind speed validation Errors (Kong et al., 2015).

Model	RMSE	MAE	MAPE
1-RSVM model	0.9087	0.6313	5.2658
2-RSVM model	0.8268	0.5271	4.1208
3-RSVM model	0.6798	0.4900	2.8990
4-RSVM model	0.8080	0.5855	6.5863

indicated that the developed C-LSSVM-PSOGSA model significantly improved the accuracy of forecasting one-day-ahead wind speed on a daily scale.

Hu et al. (2015) proposed a hybrid modeling approach, EWT-CSA-LSSVM, based on LSSVM for one-step ahead and multi-step ahead wind speed prediction in Hebei Province in China. Comparison between AR, CSA-LSSVM (linear kernel), persistence method and EWT-CSA-LSSVM (linear kernel) models was carried out via RMSE, MAE and MAPE indices to exhibit capability of the proposed model. The simulation results for datasets in the literature proved that the suggested forecasting method yielded better predictions in comparison with the others.

Wang, J. et al. (2015) used a hybrid model based on SVR to forecast short-term wind speed. RBF kernel function was considered and different optimization algorithms, such as GA, PSO and COA, were applied to optimize its parameters and compare the models by applying to wind speed prediction at the Shandong wind farm in China. The prediction horizon was 15 min (one-step-ahead), 30 min (two-step-ahead), 45 min (three-step-ahead) and 1 h ahead (four-step-ahead). It was found that the COA-SVR method was statistically robust in multi-step-ahead prediction and could be applied to practical wind farm applications.

Sun et al. (2015) adopted a hybrid model, fast ensemble empirical model decomposition (FEEMD)-bat algorithm (BA)-LSSVM, to forecast wind speed. A dataset was collected from the Rose Camp wind farm in Inner Mongolia in China, which were measured every 20 min from 1 January 2011 to 3 February 2011. In order to compare and show the ability of the model, three statistical tests, including MAE, MAPE and RMSE, were carried out. The comparison between the aforementioned model and the others concluded the robustness of the hybrid model.

Gani et al. (2016) developed an SVM-FFA to forecast wind speed distribution for both daily and monthly scales by taking wind speed data as the only input. 3-hourly wind speed data were applied to earn daily data and then the required monthly data obtained from daily wind data for each specific month. Comparing the aforementioned model results against thoes of single SVM and ANN via MAPE, RMSE and R statistical error tests exhibited robustness of the model even the performance for the daily prediction was not as good as monthly one.

Santamaría-Bonfil et al. (2016) developed a hybrid methodology based on SVR for forecasting wind speed to show how the aforementioned model outperforms AR, ARMA and ARIMA models as well as ordinary least squares method. To determine a quantitative measure of robustness of the developed approach, MAE, mean bias error (MBE), RMSE and mean absolute scaled error were utilized. GA was used as optimizer to precisely select SVM parameters. Experimental data from the Mexican Wind Energy Technology Center were applied to develop the proposed model. Comparison illustrated ability and integrity of applying the hybrid methodology based on SVR as an accurate model for predicting wind speed versus other artificial intelligence models.

Jiang et al. (2017) proposed a new short-term wind speed forecasting model based on SVM by considering the wind speed fluctuation information as additional information in forecasting wind speed. The wind speed data were collected from a wind farm in Peng Lai in China. To test the effectiveness of the proposed model, the chosen wind speed data were recorded every 10 min. The suitable inputs were fed into the SVM model that was optimized by cuckoo search algorithm. The proposed model presented better forecasts than ARIMA and the other SVMs optimized by PSO.

Jiang and Huang (2017) adopted the feature selection and error correction in the real-time decomposition-based estimating model to increase the estimation accuracy. Besides, LSSVM was used to establish the one-step ahead predicting models; finally, the hybrid LSSVM and generalized auto-regressive conditionally heteroscedastic model was proposed. According to two sets of measured data in Colorado and Minnesota, the results of this study indicated that, compared with other involved methods, the proposed hybrid method had satisfactory performance in both accuracy and stability.

6.2. SVM for wind power

Liu et al. (2009) used a piecewise SVM for short-term wind power prediction using a dataset from a wind farm in north China. In this article, wind speed and wind power were selected as the inputs and effects of climate, temperature and atmospheric pressure were neglected. The comparison between estimations by the model and single SVM via average relative error (ARE) and RMSE exhibited that the precision of piecewise SVM model had positive significance in forecasting short-term wind power.

Kramer and Gieseke (2011) used an SVR using RBF kernel function for wind energy prediction in 10 min to 6 h range using a dataset from 2004 to 2006 from NREL western wind resource, which were measured every 10 min. In this study the authors focused on wind grid points in the wind park of Tehachapi in California in USA. At the end of this analysis, it was concluded that SVR is a successful method for forecasting wind energy only based on wind measurements from windmills, in particular without further meteorological data or weather forecasts.

Giorgi et al. (2014) developed two intelligent methods, such as ANN and LSSVM with wavelet decomposition, to estimate wind power. The data to develop the model were collected from a wind farm located in South Italy for a period of 5 years with a recorded measurement of every 10 min. It was found that methods based on LSSVM performed better than ANN.

Najeebullah et al. (2015) applied a hybrid model based on SVM, SVR-LMNN-BFGSNN-BRNN, called machine learning based short term wind power prediction (MT-STWP), to predict short-term wind power. In this analysis, performance of the proposed model was compared with those of SVR-LMNN, SVR-LMNN-BFGSNN as well as Grassi model and Nima Amjady model. The dataset was collected from School of Agriculture and Natural Resources, Morrisville State College in duration of 2007 till 2011. The experimental information comprised the wind speed, relative humidity, temperature and wind power. The training data included 78% of data set and the remaining 22% were dedicated for testing the model. The error analysis demonstrated that MT-STWP had better performance than the Grassi and Nima models, as shown in Fig. 7.

Yuan et al. (2015) developed an LSSVM-GSA model for short-term wind power prediction based on wind speed and wind direction input vectors. In this analysis, the authors evaluated

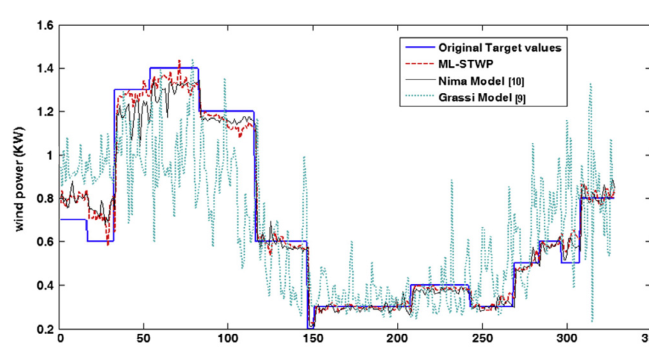

Fig. 7. Comparison of wind power prediction by various models (Najeebullah et al., 2015).

different kernel functions to earn the best one. The comparison analysis, covering LSSVM-GSA using ERBF and five different forecasting models, i.e. back propagation (BP) neural network, single SVM, LSSVM and SVM-GSA, demonstrated that the proposed model had considerably more accurate results than the other models, in which the LSSVM-GSA error analysis presented the lowest percentage, as shown in Fig. 8.

The SVM-Enhanced Markov model was developed and proposed by Yang et al. (2015) for forecasting short-term wind power. In this study, the authors collected data samples of a real wind frame that were measured every 10 min in the years of 2009 and 2010. The data of year 2009 were implemented to train the Markov chains and the SVM classifiers, and the data of year 2010 were utilized to test the forecast accuracy of the proposed model. The authors claimed that the proposed model was able enough to considerably enhance the precision of forecasting short-term wind power.

Wu and Peng (2016) developed a novel hybrid model based on LSSVM to predict wind power by combining LSSVM, ensemble empirical mode decomposition (EEMD), principal component analysis (PCA) and bat algorithm (BA). 301 data samples between 1 January 2015 and 28 October 2015 were collected from a wind frame that was located in Hebei province in China. Various statistical criteria, such as RMSE, MAE and MAPE, were chosen to assess the model ability. Results of the comparison with the other models proved superiority of EEMD-PCA-LSSVM-BA model.

Yuan et al. (2017) offered a hybrid autoregressive fractionally integrated moving average and LSSVM model to estimate short-term wind power. Firstly, the authors used the autocorrelation function analysis to detect the long memory characteristics of wind power series, and the autoregressive fractionally integrated moving average method was implemented to predict linear component of wind power series. After that, the LSSVM model was established to predict nonlinear component of wind power series. Compared with other models, the simulation results showed that the suggested hybrid model had the lowest values of RMSE, MAPE and MAE.

Liu et al. (2017b) offered a hybrid estimating model termed as orthogonal test and SVM (OT-SVM). A novel factor analysis method was developed based on the theory of OT, and used to select the optimal inputs of SVM approach. The effectiveness of the OT-SVM was evaluated in three wind farms in China, and the forecasting results were compared with other benchmark methods. The results proved that the suggested OT-SVM had the highest robustness covering different input numbers and time resolutions.

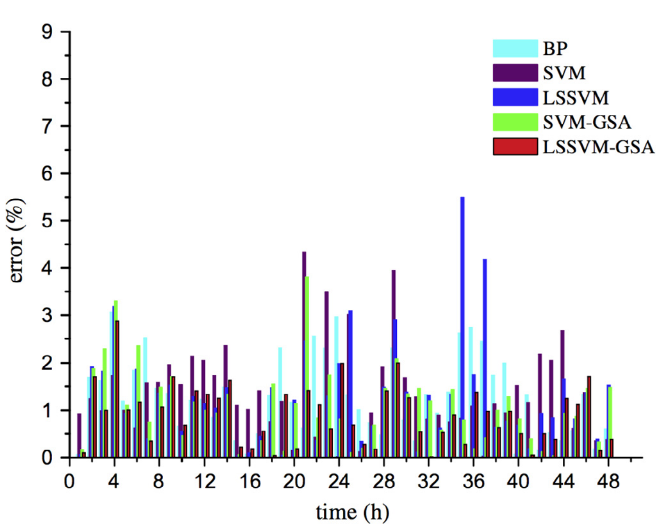

Fig. 8. Absolute error distribution curve for different models (Yuan et al., 2015).

Park and Hur (2017) proposed a method for estimating wind power outputs using an SVM based on multi-variable regression. The proposed model was verified with experimental data from a wind turbine located in Jeju Island in Korea. After training, the authors used the model to estimate wind power over the next 24 h. The results showed that estimated values using the SVM based on multi-variable regression was more accurate than the SVM based on single-variable regression.

6.3. SVM for wind direction

Two methods for estimating short-term wind direction based on SVR using RBF as kernel function and ANN were developed by Tagliaferri et al. (2015). The authors considered input and output parameters as the wind direction measurements in the past minutes and the wind direction for the next 2 min, respectively. The data samples used for this work consisted of registrations collected during the 34th America's Cup in San Francisco. The authors used the last 100 min to test the performance of the ANN and SVR methods, and the rest of the data samples to train both models. However, comparison of the models via MAE and mean effectiveness index proved that the SVR had a better capability in terms of accuracy and computational time.

7. Future work

In most of the cited literature pertaining to accuracy of forecasting methods, only a particular location is considered. The recommendations presented and implementations proposed were based on statistical tests in a specific region only. It is recommended to test a particular wind and solar energy model in different geographical regions with different overall pattern to compare results for a better accuracy.

In spite of the advancements made thus far, it seems that the solar air heater systems, solar insolation, and wind directions have been received scant attention by researchers. More attention is needed for these parameters due to its importance in developing robust and accurate models.

It can be seen that LSSVM with a high generalization capability not only uses merits of SVM but also solves a set of linear equations that is easier to solve compared to quadratic programming language. The favorable performance of the aforementioned approach in conducting research is really attractive due to its merits. It should be introduced as the promising model, which needs more investigation in the research areas and there are many windows to discover about the remarkable model in the fields of forecasting problems.

Moreover, precise selection of SVMs parameters has a huge effect on their accuracies. Gradient descent and grid search algorithms, which can be considered as the conventional methods, could not be selected as the best choice due to their drawbacks, such as computational complexity. Additionally, PSO and harmony search (HS) algorithms performances have been proved to be poor and might lead to slow convergence speed. Combination of LSSVM with GA could be considered as one of the best approaches, which has been successfully applied in the fields of chemical (Hemmati-Sarapardeh et al., 2014) and air-conditioning (Zendehboudi, 2016) systems. Also, the performance of BA to search optimal parameters could not be neglected, since it has shown excellent results. Moreover, combination of LSSVM with FFA should be tested and further researched due to the merits of FFA in solving optimization problems.

Furthermore, WT algorithm, which is a multi-resolution data-processing tool for non-stationary input signals, could be replaced by fast ensemble empirical model decomposition algorithm to boost the real-time compuatational performance. This algorithm was proposed in 2014 and has shown considerable performance improvements.

Additionally, experimental data always come with uncertainties that bring about lower accuracy when a model is developed. Therefore, outlier detection should be performed to evaluate data samples and detect those that are far from bulk of dataset. The outlaying observations or outliers appear due to different elements, such as changes in system behavior, mechanical faults, instrument error and human error. In this case, these observations have the potential to negatively affect the overall predictive ability of a fitted model. Their detection using a mathematical approach can extremely improve the efficiency and integrity of a constructed model.

8. Concluding remarks

The global increase in solar and wind energy penetration in the last one and half decade has necessitated the need of accurate and quick resource predictions. These accurate predictions will ensure optimum utilization of sustainable and renewable sources of energy and thereby reduce the harmful effects of the use of fossil fuel resources. This paper reviewed application of SVM modeling approach in the fields of solar and wind energy. This model has attracted attention of many scholars worldwide and been widely utilized due to its reliability and accuracy in prediction. However, on the basis of the reviewed related articles in the both parts of wind and solar, the following conclusions can be drawn:

I. Developing a globally consistent model for wind and solar energy prediction is not feasible. Both solar and wind resources exhibit variable patterns based on a variety of factors.
II. By utilizing various statistical error tests, ANN, ANFIS, RBFNN, ARIMA, LES, SES, RW, etc. models do not exhibit bright integrity in forecasting of both solar and wind energy in most of the locations; however, in general, implementation of SVM could be an effective and precise modeling approach, especially in short-term forecasting.
III. Suitable hyper parameters in the model have a huge effect on the accuracy of outputs; consequently, the forecasting precision and ability of SVM is better in comparison with the conventional models and gives encouraging results when it integrates and introduces as a hybrid model. However, a hybrid model may provide slow response to new forecasting data.
IV. SVM modeling approach enjoys certain privileges like ease in calculation with high efficiency, simple-to-use and considerably less computational time and cost, which make it attractive to many scholars to use it in different problems.

However, further investigations are ongoing to develop powerful hybrid models in order to enhance speed, precision and ability of forecasting using SVM methods.

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